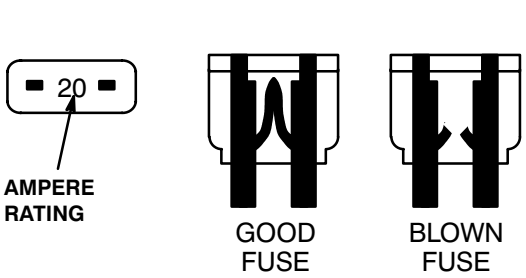
11-1 FUSE PANEL/CIRCUIT PROTECTION

CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fusible links, maxi-fuses, circuit breakers, or a combination of these devices.

BLADE TYPE FUSE

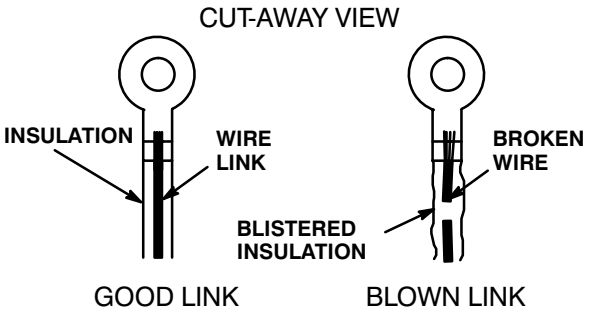


Blade type fuses have a transparent, color-coded, plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code :

BLADE FUSE COLOR-CODING	
AMPERE RATING	**HOUSING COLOR**
2	Gray
4	Pink
5	Tan
7.5	Brown
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

FUSIBLE LINK



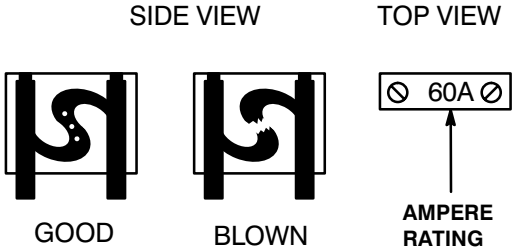
Fusible links are short lengths of wire that are smaller in diameter than the wires they are protecting. Fusible link wire is covered with a special thick, non-flammable insulation. An overload condition causes the insulation to blister. If the overload condition continues, the wire link will melt. To check a fusible link, look for blistered insulation. If the insulation is okay, pull lightly on the wire; If the fusible link stretches, the wire has melted.

When replacing fusible links, first cut the protected wire where it is connected to the fusible link. Then, tightly crimp or solder the new link to the protected wire and cover the joint with insulating tape.

Fusible links are often identified by color coding of the insulation, as shown here:

FUSIBLE LINK COLOR-CODING	
WIRE LINK SIZE	**INSULATION COLOR**
20 GA	Blue
18 GA	Brown or Red
16 GA	Black or Orange
14 GA	Green
12 GA	Gray

MAXI-FUSE



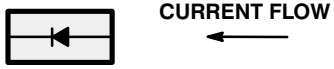
Maxi-fuses have a transparent, color-coded plastic housing. To check a maxi-fuse, look at the fuse element through the side of the housing.

To replace a maxi-fuse, pull it from the fuse box or panel. Always replace a blown maxi-fuse with a new one having the same ampere rating.

The ampere rating of a maxi-fuse can also be determined by following the color-code:

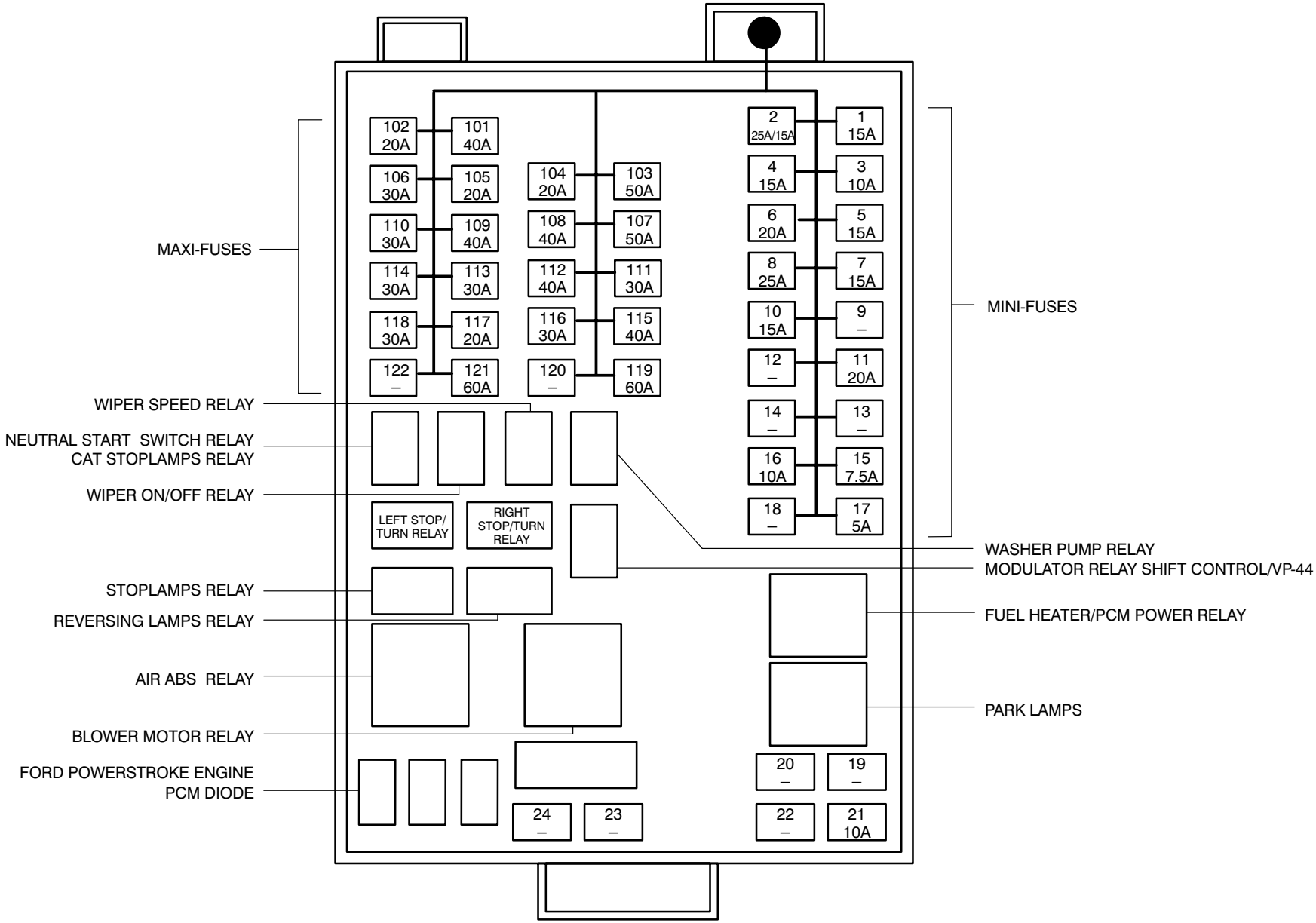
MAXI-FUSE COLOR-CODING	
AMPERE RATING	**HOUSING COLOR**
30	Light Green
40	Amber
50	Red
60	Blue

DIODE



Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

BATTERY JUNCTION BOX



11-3 FUSE PANEL/CIRCUIT PROTECTION

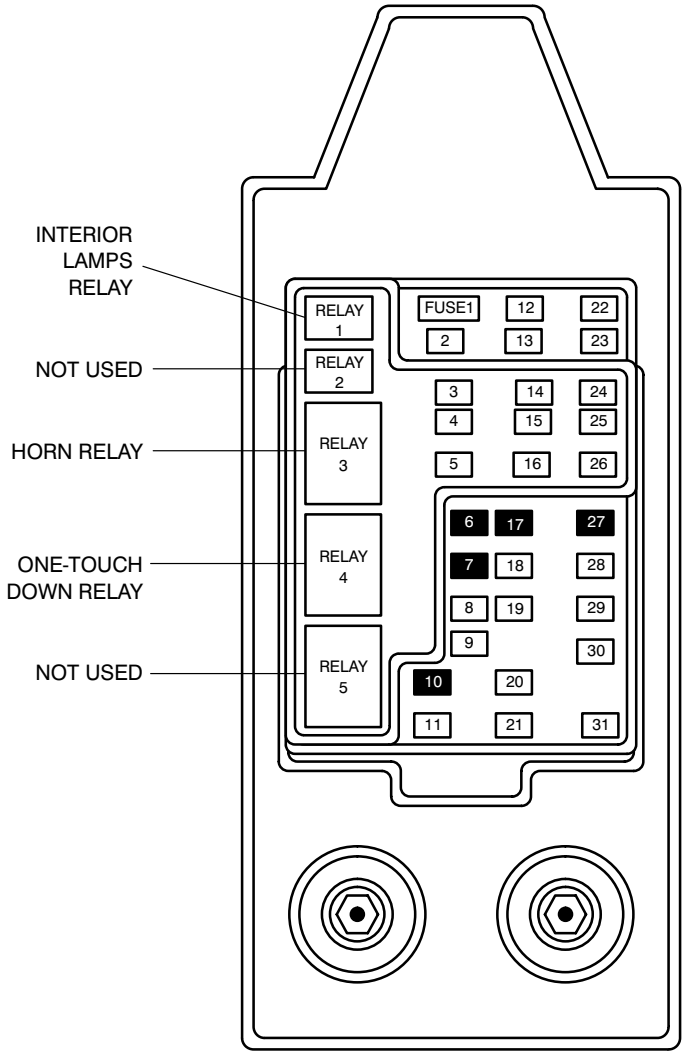
*** SUPERCREWZER**

BATTERY JUNCTION BOX

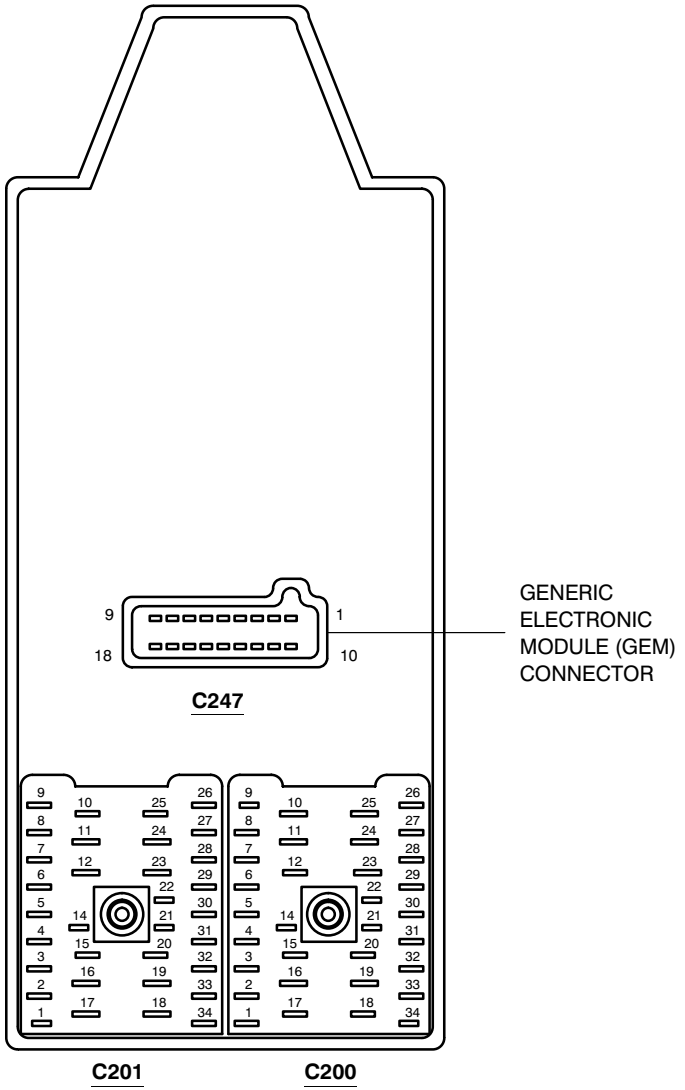
Fuse Position	Amps	Circuits Protected
1	15	Main Light Switch, Park Lamps, Marker Lamps, Park Lamps Relay, Central Junction Box Fuse #18
2	25	VP-44 Relay (Cummins), Cummins Engine Control Module
*2	15	Fog Lamps Relay
3	10	Generator
4	15	Washer Pump Relay, Washer Pump Motor
5	15	Engine Controls
6	20	Engine Controls
7	15	Stoplamp Switches
8	25	Fuel Heater Relay (Cat)
9	–	Not Used
10	15	Downshift Solenoid
11	20	VIM
12-14	–	Not Used
15	7.5	Trailer Adapter
16	10	Allison Electronic Control Unit (2000/2400)
*16	10	Vehicle Interface Module (VIM)
*17	5	Radio
17-20	–	Not Used
21	10	Hydromax Motor
22-24	–	Not Used
101	40 Maxi	Air ABS Relay, Air ABS Module
102	20 Maxi	Ignition Switch to Trailer Adapter
103	50 Maxi	Ignition Switch, Central Junction Box Fuse # 8, 9, 11, 19, 20, 23, 24, 25, 29, 30, 31

Fuse Position	Amps	Circuits Protected
104	20 Maxi	Power Point
105	20 Maxi	Door Lock Switches
106	30 Maxi	Main Light Switch, Multifunction Switch, Central Junction Box Fuse #16, 26, 28, Headlamps, DRL Relays
107	50 Maxi	Central Junction Box Fuse #15, GEM, Interior Lamps Relay
108	40 Maxi	Fuel Heater Relay
109	40 Maxi	Window Switches, Flasher Relay
110	–	Not Used
*110	30	Power Seats
111	30 Maxi	Park Lamps Relay, Park Lamps
112	40 Maxi	Blower Motor Relay, Blower Motor
113,114	–	Not Used
*113	30	Heated Seats
*114	30	E-Brakes
115	40 Maxi	Ignition Switch, Central Junction Box Fuse #5, 8, 9, 11, 21
116	30 Maxi	Left and Right Stop/Turn Relays, Stop and Reversing Lamps Relays
117	40 Maxi	Ford Powerstroke Engine PCM Power Relay
118	30 Maxi	Ford Powerstroke Diesel
119,120	60 Maxi	ABS Module Power
*119,120	60	Air Brake, Trailer Tow Fuse Block
121,122	60 Maxi	Hydromax Relay
*121,122	60	Air Brake, Trailer Tow Fuse Block

CENTRAL JUNCTION BOX



TOP VIEW



BOTTOM VIEW

11-5 FUSE PANEL/CIRCUIT PROTECTION

*** SUPERCREWZER**

CENTRAL JUNCTION BOX

Fuse Position	Amps	Circuits Protected
1	20	Horn Relay
2	15	Flasher Relay
3	20	Cigar Lighter
4	10	Data Link Connector (DLC), Engine Diagnostic Connector
5	15	Blend Door Actuator, Mode Selector Switch, Reversing Lamps Switch, DRL
*5	15	Heated Seat Module, On/Off Fan Clutch, Warning Chime Module
6	–	Not Used
7	–	Not Used
8	5	GEM, Radio
9	5	Power Window, Door Lock Switches, Main Light Switch
*9	5	Trailer Hookup Lamps
10	–	Not Used
11	30	Windshield Wiper Relays, Wiper Motor
12	10	Stoplamp Switch
13	20	Instrument Cluster, Radio, Powertrain Control Module (PCM)
14	10	Interior Lamps
*14	10	Running Board Lamps
15	10	GEM, Interior Lamps Relay
*15	10	Map Lamps, Glove Compartment Lamps, Power Mirrors, Warning Chime Module
16	15	High Beam Headlamps and Indicator

Fuse Position	Amps	Circuits Protected
17	–	Not Used
18	5	Main Light Switch
19	15	Engine Controls, Instrument Cluster
20	15	GEM, Clutch Interlock Switch, VIM, Neutral Switch
21	10	DRL
22	15	Two-Speed Axle (Mexico)
*22	15	Exhaust Brake, Heated Mirrors, Rear Air Suspension
23	10	Flasher Relay
24	15	Air Brakes, ABS Module, Fuel Heater Relay, Vacuum Pump Assembly
25	10	Blower Motor Relay (Cummins)/Function Selector Switch (CAT/Ford Powerstroke Diesel)
26	10	Right Headlamp
27	–	Not Used
28	10	Left Headlamp
29	10	GEM, Idle Validation Switch, Instrument Cluster, Clutch Pedal Position Switch
*29	10	Overhead Console, Remote Keyless Entry Module, Trax (Fuel) Module
30	30	Fuel Heater, PCM Diode (Ford Powerstroke Diesel)
31	15	Vehicle Interface Module (CAT)/Neutral Switch (Ford Powerstroke Diesel)